

Roll No.....

ODD END SEMESTER EXAMINATION DECEMBER - 2025

Name of the Program: **B. Tech. Petroleum Engineering**

Semester: **III**

Name of the Course: **General Geology**

Course Code: **TPE 302**

Time: **3 Hours**

Maximum Marks: **100**

Note:

- (i) All questions are compulsory.
- (ii) Answer any **two** sub questions among **a, b & c** in each main question.
- (iii) Total marks for each main question are **twenty**.
- (iv) Each question carries **10 Marks**

Q1	(20 marks)	
(a)	Explain the importance of physical properties in mineral identification. Briefly outline the key properties and their practical application in field studies.	CO1 CO2
(b)	Distinguish between weathering and erosion. What are the main processes that occur during weathering?	
(c)	Explain the rock cycle and discuss the different stages and processes involved.	
Q2	(20 marks)	
(a)	Explain the internal structure of the Earth. Describe the characteristics of each layer and illustrate them with a neat labelled diagram.	CO2 CO3
(b)	Differentiate between i) Fracture and Cleavage ii) Dike and Sill	
(c)	Define plate tectonics. Discuss the various types of plate boundaries and how they differ from one another.	
Q3	(20 marks)	
(a)	What do you mean by texture? Discuss various texture associated with igneous rocks.	CO3 CO4
(b)	Explain the classification system and the significance of sedimentary rocks.	
(c)	What is a rock? Explain the properties that distinguish igneous, sedimentary, and metamorphic rocks from one another.	
Q4	(20 marks)	
(a)	Define an earthquake and describe the geological processes responsible for its occurrence. Explain the different types of seismic waves produced and how they vary in speed and motion.	CO4 CO5
(b)	Discuss the recording and measurement of earthquakes. What types of structural and environmental damage can occur during strong earthquakes?	
(c)	What are volcanoes? Describe the different types of volcanoes.	
Q5	(20 marks)	
(a)	Discuss the mechanism of folding in rocks under compressional forces. Classify the different types of folds with labeled diagrams.	CO5 CO6
(b)	What is a fault? Describe its structure and mechanics. Classify the various types of faults and illustrate your answer with suitable sketches.	
(c)	Define a joint in geological terms. Differentiate between joints and faults, and explain the classification of joints based on their origin and appearance.	

